7D_1 is the Case where the curled geometry is the product of 3 circles, $S^1xS^1xS^1$.

Notation and Setup:

Starting Metric for this Case: $g_{00}=\frac{6\,\dot{a}[t]^2}{-6\,k+\Lambda\,a[t]^2}$, $g_{11}=-\frac{a[t]^2}{\mathcal{D}_4[r]\,\mathcal{D}_5[r]\,\mathcal{D}_6[r]}$, $g_{22}=-r^2\,a[t]^2$, $g_{33}=-r^2\,a[t]^2\,\text{Sin}[\theta]^2$, $g_{44}=-a[t]^2\,p_4^2\,\mathcal{D}_4[r]$, $g_{55}=-a[t]^2\,p_5^2\,\mathcal{D}_5[r]$, $g_{66}=-a[t]^2\,p_6^2\,\mathcal{D}_6[r]$, a(t) not constant

Corresponding Line Element

$$ds^2 = \frac{6\,\dot{a}[t]^2}{-6\,k+\Lambda\,a[t]^2}\,dt^2 - \frac{a[t]^2}{\mathcal{D}_4[r]\,\mathcal{D}_5[r]\,\mathcal{D}_6[r]}\,dr^2 - r^2\,a[t]^2\,d\theta^2 - r^2\,a[t]^2\,\text{Sin}[\theta]^2\,d\phi^2 - a[t]^2\,p_4^2\,\mathcal{D}_4[r]\,d\psi_4^2 - a[t]^2\,p_5^2\,\mathcal{D}_5[r]\,d\psi_5^2 - a[t]^2\,p_6^2\,\mathcal{D}_6[r]\,d\psi_6^2$$

Output Display Simplification - Parameters are sometimes suppressed
Prime (if any) means derivative wrt r.
Dot (if any) means derivative wrt t.
Examples: **A = A(t,r)**, **$A'=\partial_r A(t,r)$** and **$\dot{A}=\partial_t A(t,r)$**

Lambda Vacuum Equations: The equations are $R_{\mu\nu} = \Lambda\, g_{\mu\nu}$. They are expressed as $\frac{R_{\mu\nu}}{g_{\mu\nu}} = \Lambda$ when $g_{\mu\nu} \neq 0$, and $R_{\mu\nu} = 0$ when $g_{\mu\nu} = 0$. The $g_{\mu\nu}$ matrix used is a diagonal matrix. Since off-diagonal $R_{\mu\nu}$ equations rarely appear, almost all the equations are expressed by dividing out $g_{\mu\nu}$.

GRW Equations: The exact solution to the Λ vacuum field equations that has (1) C = 1, (2) a(t) is a free parameter, and (3) parameter $\mathcal{A}$ such that $\mathcal{A}(t) = c^2$ is the Friedmann Continuity Equation (which generates the a(t) formula).

FG-GRW Equations: An exact solution to the Λ vacuum field equations that has (1) free parameters a and C, and (2) parameter $\mathcal{A}$ such that $\mathcal{A}(t) = c^2$ is the Friedmann Continuity Equation (which generates the a(t) formula).

a: GRW Equations

As usual, we seek assumptions that can simplify the ODEs.

The initial assumptions for A and C follow an established pattern:

- $A(t,r) = \mathcal{A}(t) = \frac{6\,\dot{a}[t]^2}{-6\,k+\Lambda\,a[t]^2}$, where curvature k is a free parameter
- C(r) = 1 (standard in GRW equations)

The more difficult issue is B:

- In 5D there is just one circle and we showed that $B = \frac{1}{\mathcal{D}_4}$.
- In 6D_1 there are two circles, and we assumed that $B = \frac{1}{\mathcal{D}_4\,\mathcal{D}_5}$.
- In 7D_1 we have three circles, so it may be reasonable to assume
  - ✓ $B = \frac{1}{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6}$.
  - ✓ This is the simplest assumption that treats all three circles alike.
  - ✓ Moreover, it keeps the volume element of the (r,ψ_4,ψ_5,ψ_6) subspace equal to its flat-space value: $\sqrt{B\,\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6}=1$.

Summary of assumptions:

- $A(t,r) = \mathcal{A}(t) = \frac{6\,\dot{a}[t]^2}{-6\,k+\Lambda\,a[t]^2}$, where curvature k is a free parameter,
- $B(r) = \frac{1}{\mathcal{D}_4[r]\,\mathcal{D}_5[r]\,\mathcal{D}_6[r]}$,
- $C(r) = 1$,
- $\mathcal{D}_4(r)$ - $\mathcal{D}_6(r)$ are free parameters.

The initial Lambda field equations have three unknowns, $\mathcal{D}_4$ - $\mathcal{D}_6$:

R_{00}: $\Lambda = \Lambda$

R_{01}: $0 = 0$

R_{11}: $-\frac{5\,k}{a^2}+\Lambda-\frac{\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4'}{a^2\,r}-\frac{\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5'}{a^2\,r}-\frac{\mathcal{D}_6\,\mathcal{D}_4'\,\mathcal{D}_5'}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6'}{a^2\,r}-\frac{\mathcal{D}_5\,\mathcal{D}_4'\,\mathcal{D}_6'}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_5'\,\mathcal{D}_6'}{2\,a^2}-\frac{\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4''}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5''}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6''}{2\,a^2} = \Lambda$

R_{22}: $-\frac{5\,k}{a^2}+\frac{1}{a^2\,r^2}+\Lambda-\frac{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6}{a^2\,r^2}-\frac{\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4'}{a^2\,r}-\frac{\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5'}{a^2\,r}-\frac{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6'}{a^2\,r} = \Lambda$

R_{33}: $-\frac{5\,k}{a^2}+\frac{1}{a^2\,r^2}+\Lambda-\frac{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6}{a^2\,r^2}-\frac{\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4'}{a^2\,r}-\frac{\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5'}{a^2\,r}-\frac{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6'}{a^2\,r} = \Lambda$

R_{44}: $-\frac{5\,k}{a^2}+\Lambda-\frac{\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4'}{a^2\,r}-\frac{\mathcal{D}_6\,\mathcal{D}_4'\,\mathcal{D}_5'}{2\,a^2}-\frac{\mathcal{D}_5\,\mathcal{D}_4'\,\mathcal{D}_6'}{2\,a^2}-\frac{\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4''}{2\,a^2} = \Lambda$

R_{55}: $-\frac{5\,k}{a^2}+\Lambda-\frac{\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5'}{a^2\,r}-\frac{\mathcal{D}_6\,\mathcal{D}_4'\,\mathcal{D}_5'}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_5'\,\mathcal{D}_6'}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5''}{2\,a^2} = \Lambda$

R_{66}: $-\frac{5\,k}{a^2}+\Lambda-\frac{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6'}{a^2\,r}-\frac{\mathcal{D}_5\,\mathcal{D}_4'\,\mathcal{D}_6'}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_5'\,\mathcal{D}_6'}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6''}{2\,a^2} = \Lambda$

R_{33} is redundant, the same as R_{22}.

R_{00} and R_{01} are identities due to the $\mathcal{A}[t]$ formula.

That leaves 5 equations:

$$R_{11}: \quad -\frac{5\,k}{a^2}+\Lambda-\frac{\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4'}{a^2\,r}-\frac{\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5'}{a^2\,r}-\frac{\mathcal{D}_6\,\mathcal{D}_4'\,\mathcal{D}_5'}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6'}{a^2\,r}-\frac{\mathcal{D}_5\,\mathcal{D}_4'\,\mathcal{D}_6'}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_5'\,\mathcal{D}_6'}{2\,a^2}-\frac{\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4''}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5''}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6''}{2\,a^2}=\Lambda$$

$$R_{22}: \quad -\frac{5\,k}{a^2}+\frac{1}{a^2\,r^2}+\Lambda-\frac{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6}{a^2\,r^2}-\frac{\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4'}{a^2\,r}-\frac{\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5'}{a^2\,r}-\frac{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6'}{a^2\,r}=\Lambda$$

$$R_{44}: \quad -\frac{5\,k}{a^2}+\Lambda-\frac{\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4'}{a^2\,r}-\frac{\mathcal{D}_6\,\mathcal{D}_4'\,\mathcal{D}_5'}{2\,a^2}-\frac{\mathcal{D}_5\,\mathcal{D}_4'\,\mathcal{D}_6'}{2\,a^2}-\frac{\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4''}{2\,a^2}=\Lambda$$

$$R_{55}: \quad -\frac{5\,k}{a^2}+\Lambda-\frac{\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5'}{a^2\,r}-\frac{\mathcal{D}_6\,\mathcal{D}_4'\,\mathcal{D}_5'}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_5'\,\mathcal{D}_6'}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5''}{2\,a^2}=\Lambda$$

$$R_{66}: \quad -\frac{5\,k}{a^2}+\Lambda-\frac{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6'}{a^2\,r}-\frac{\mathcal{D}_5\,\mathcal{D}_4'\,\mathcal{D}_6'}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_5'\,\mathcal{D}_6'}{2\,a^2}-\frac{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6''}{2\,a^2}=\Lambda$$

Each equation can be simplified by subtracting Λ, putting the remainder over a common denominator, and setting the numerator to zero:

$$R_{11}: \quad -10\,k\,r-2\,\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4'-2\,\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5'-r\,\mathcal{D}_6\,\mathcal{D}_4'\,\mathcal{D}_5'-2\,\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6'-r\,\mathcal{D}_5\,\mathcal{D}_4'\,\mathcal{D}_6'-r\,\mathcal{D}_4\,\mathcal{D}_5'\,\mathcal{D}_6'-r\,\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4''-r\,\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5''-r\,\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6''=0$$

$$R_{22}: \quad 1-5\,k\,r^2-\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6-r\,\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4'-r\,\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5'-r\,\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6'=0$$

$$R_{44}: \quad -10\,k\,r-2\,\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4'-r\,\mathcal{D}_6\,\mathcal{D}_4'\,\mathcal{D}_5'-r\,\mathcal{D}_5\,\mathcal{D}_4'\,\mathcal{D}_6'-r\,\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4''=0$$

$$R_{55}: \quad -10\,k\,r-2\,\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5'-r\,\mathcal{D}_6\,\mathcal{D}_4'\,\mathcal{D}_5'-r\,\mathcal{D}_4\,\mathcal{D}_5'\,\mathcal{D}_6'-r\,\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5''=0$$

$$R_{66}: \quad -10\,k\,r-2\,\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6'-r\,\mathcal{D}_5\,\mathcal{D}_4'\,\mathcal{D}_6'-r\,\mathcal{D}_4\,\mathcal{D}_5'\,\mathcal{D}_6'-r\,\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6''=0$$

R_{44}–R_{66} can each be expressed as an exact derivative

$$R_{44}: \quad (r^2\,\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4')'=-10\,k\,r^2.$$

$$R_{55}: \quad (r^2\,\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5')'=-10\,k\,r^2.$$

$$R_{66}: \quad (r^2\,\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6')'=-10\,k\,r^2.$$

Integrating once gives

$$R_{44}: \quad r^2\,\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4'=-\frac{10\,k\,r^3}{3}+\zeta_4 \quad \text{where } \zeta_4 \text{ is a constant of integration}$$

$$R_{55}: \quad r^2\,\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5'=-\frac{10\,k\,r^3}{3}+\zeta_5 \quad \text{where } \zeta_5 \text{ is a constant of integration}$$

$$R_{66}: \quad r^2\,\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6'=-\frac{10\,k\,r^3}{3}+\zeta_6 \quad \text{where } \zeta_6 \text{ is a constant of integration}$$

Define $Q(r)=\mathcal{D}_4[r]\,\mathcal{D}_5[r]\,\mathcal{D}_6[r]$.

$$\Rightarrow \quad Q'=\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4'+\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5'+\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6'$$

Using $Q(r)$, R_{22} is also exact:

R_{22}: $(rQ)' = \mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6 + r\,\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4' + r\,\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5' + r\,\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6' = 1 - 5\,k\,r^2$

Integrating once gives

R_{22}: $Q = 1 - \frac{5\,k\,r^2}{3} + \frac{\chi}{r}$ where χ is a constant of integration

$\Rightarrow \quad Q' = -\frac{10\,k\,r}{3} - \frac{\chi}{r^2}$

$\Rightarrow \quad r^2 Q' = -\frac{10\,k\,r^3}{3} - \chi$

$\Rightarrow$ $R_{44}+R_{55}+R_{66}$: $r^2 Q' = r^2\,\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4' + r^2\,\mathcal{D}_4\,\mathcal{D}_6\,\mathcal{D}_5' + r^2\,\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6' = -10\,k\,r^3 + \zeta_4 + \zeta_5 + \zeta_6$

Setting these two expressions for $r^2 Q'$ equal to each other:

$$-\frac{10\,k\,r^3}{3} - \chi = -10\,k\,r^3 + \zeta_4 + \zeta_5 + \zeta_6$$

or $20\,k\,r^3 - 3\,(\chi + \zeta_4 + \zeta_5 + \zeta_6) = 0$

$\Rightarrow \quad k = 0$

and $\chi = -\zeta_4 - \zeta_5 - \zeta_6$

$\Rightarrow \quad R_{22}$: $Q = 1 + \frac{\chi}{r}$

$\Rightarrow \quad R_{44}$: $r^2\,\mathcal{D}_5\,\mathcal{D}_6\,\mathcal{D}_4' = \zeta_4$

$\Rightarrow \quad \frac{\mathcal{D}_4'}{\mathcal{D}_4} = \frac{\zeta_4}{r^2\,\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6} = \frac{\zeta_4}{r^2 Q} = \frac{\zeta_4}{r\,(r+\chi)}$

$\Rightarrow \quad \text{Log}[\mathcal{D}_4] = \int \frac{\zeta_4}{r\,(r+\chi)}\,dr = \begin{cases} \frac{\zeta_4\,(\log(r)-\log(r+\chi))}{\chi} = \frac{\zeta_4 \log\left(\frac{r}{r+\chi}\right)}{\chi} & \text{if } \chi \neq 0 \\ -\frac{\zeta_4}{r} & \text{if } \chi = 0 \end{cases}$

$\Rightarrow \quad \mathcal{D}_4 = \begin{cases} \left(\frac{r}{r+\chi}\right)^{\frac{\zeta_4}{\chi}} & \text{if } \chi \neq 0 \\ e^{-\frac{\zeta_4}{r}} & \text{if } \chi = 0 \end{cases}$

and, similarly

$\Rightarrow \quad \mathcal{D}_5 = \begin{cases} \left(\frac{r}{r+\chi}\right)^{\frac{\zeta_5}{\chi}} & \text{if } \chi \neq 0 \\ e^{-\frac{\zeta_5}{r}} & \text{if } \chi = 0 \end{cases}$

$\Rightarrow \quad \mathcal{D}_6 = \begin{cases} \left(\frac{r}{r+\chi}\right)^{\frac{\zeta_6}{\chi}} & \text{if } \chi \neq 0 \\ e^{-\frac{\zeta_6}{r}} & \text{if } \chi = 0 \end{cases}$

Using these formulas, R_{44} - R_{66} are satisfied.

The numerator=0 form of these equations, above, yields

R_{44}: $0 = 0$

R_{55}: $0 = 0$

R_{66}: $0 = 0$

Only equation R_{11} remains to be satisfied.
Replacing χ by $-\zeta_4-\zeta_5-\zeta_6$, the numerator=0 version of R_{11} is

$$R_{11}: \quad -\frac{\zeta_5\,\zeta_6+\zeta_4\,(\zeta_5+\zeta_6)}{r^2\,(-r+\zeta_4+\zeta_5+\zeta_6)} = 0$$

$$\Rightarrow \quad \zeta_4\,\zeta_5+\zeta_4\,\zeta_6+\zeta_5\,\zeta_6 = 0$$

Thus, the final constraint is that only two of the three ζ_μ are free.
If we write the formula for any one of them in terms of the other two,
the final equation is satisfied and we have an exact solution. We choose

$$\zeta_4 = -\frac{\zeta_5\,\zeta_6}{\zeta_5+\zeta_6}$$

Due to this formula for ζ_4, this case has a solution with two branches:

Branch 1. $\zeta_4 = 0$:

Due to the ζ_4 formula, precisely one of $\zeta_5 = 0$ or $\zeta_6 = 0$.

The ψ_5 and ψ_6 circles are interchangeable at this point, so we set

$$\zeta_5 = 0$$

$$\Rightarrow \quad \chi = -\zeta_6 \neq 0$$

$$\Rightarrow \quad \text{Only the } \chi\neq 0 \text{ pieces of } \mathcal{D}_4,\ \mathcal{D}_5, \text{ and } \mathcal{D}_6, \text{ above, apply}$$

$$\Rightarrow \quad \mathcal{D}_4(r) = \mathcal{D}_5(r) = 1$$

$$\text{and} \quad \mathcal{D}_6(r) = 1+\frac{\chi}{r}$$

$$\Rightarrow \quad B(r) = \frac{1}{\mathcal{D}_6[r]}$$

That is, we have found a GRW solution:

Curvature $k = 0$,

$$A(t,r) = \mathcal{A}(t) = \frac{6\,\dot{a}[t]^2}{\Lambda\,a[t]^2} = c^2 \quad \text{where } \Lambda > 0,$$

$$B(r) = \frac{1}{\mathcal{D}_6[r]},$$

$$C(r) = 1,$$

$$\mathcal{D}_4(r) = \mathcal{D}_5(r) = 1,$$

$$\mathcal{D}_6(r) = 1+\frac{\chi}{r} \quad \text{where } \chi \neq 0,$$

and line element

$$ds^2 = c^2\,dt^2 - \frac{a[t]^2}{1+\frac{\chi}{r}}\,dr^2 - r^2\,a[t]^2\,d\theta^2 - r^2\,a[t]^2\,\text{Sin}[\theta]^2\,d\phi^2$$
$$- a[t]^2\,p_4^2\,d\psi_4{}^2 - a[t]^2\,p_5^2\,d\psi_5^2 - (1+\frac{\chi}{r})\,a[t]^2\,p_6^2\,d\psi_6^2$$

Observation:

This solution is the special case of the Branch 2 formula having ζ_5=0.

It is the solution selected by the KK-bubble smoothness constraint.

Branch 2. $\zeta_4 \neq 0$:

When $\chi \neq 0$,

$$\mathcal{D}_4(r) = (1+\frac{\chi}{r})^{-\frac{\zeta_4}{\chi}} \quad \text{where } \zeta_4 = -\frac{\zeta_5\,\zeta_6}{\zeta_5+\zeta_6},$$

$$\mathcal{D}_5 = \left(1+\frac{\chi}{r}\right)^{-\frac{\zeta_5}{\chi}},$$

and

$$\mathcal{D}_6 = \left(1+\frac{\chi}{r}\right)^{-\frac{\zeta_6}{\chi}}.$$

When $\chi = 0$,

$$0 = \zeta_4+\zeta_5+\zeta_6 = \zeta_5+\zeta_6-\frac{\zeta_5\,\zeta_6}{\zeta_5+\zeta_6} = \frac{\zeta_5^2+\zeta_5\,\zeta_6+\zeta_6^2}{\zeta_5+\zeta_6}$$
$$\Rightarrow \quad \zeta_5^2+\zeta_5\,\zeta_6+\zeta_6^2 = 0$$

whose only real solution is

$$\zeta_5 = \zeta_6 = 0 \;\Rightarrow\; \zeta_4 = 0 \;\Rightarrow\; \mathcal{D}_4 = \mathcal{D}_5 = \mathcal{D}_6 = 1,$$

the trivial metric (which we ignore).

The Branch 2 (non-trivial) GRW exact solution to the Λ field equations has zero curvature, k=0, and

$$A(t,r) = \mathcal{A}(t) = \frac{6\,\dot{a}[t]^2}{\Lambda\,a[t]^2} = c^2 \quad \text{where } \Lambda > 0,$$

$$B(r) = \frac{1}{\mathcal{D}_4[r]\,\mathcal{D}_5[r]\,\mathcal{D}_6[r]} = \frac{1}{1+\frac{\chi}{r}} \quad \text{where } \chi \neq 0,$$

$$C(r) = 1,$$

$$\chi = -(\zeta_4+\zeta_5+\zeta_6),$$

$$\zeta_4 = -\frac{\zeta_5\,\zeta_6}{\zeta_5+\zeta_6} \quad \text{where } \zeta_5\neq 0,\ \zeta_6\neq 0, \text{ and } \zeta_5+\zeta_6\neq 0\ ,$$

$$\mathcal{D}_4(r) = \left(1+\frac{\chi}{r}\right)^{\frac{\zeta_5\,\zeta_6}{\chi\,(\zeta_5+\zeta_6)}},$$

$$\mathcal{D}_5(r) = \left(1+\frac{\chi}{r}\right)^{-\frac{\zeta_5}{\chi}},$$

$$\mathcal{D}_6(r) = \left(1+\frac{\chi}{r}\right)^{-\frac{\zeta_6}{\chi}},$$

and line element

$$ds^2 = c^2\,dt^2 - \frac{a[t]^2}{1+\frac{\chi}{r}}\,dr^2 - r^2\,a[t]^2\,d\Theta^2 - r^2\,a[t]^2\,\text{Sin}[\Theta]^2\,d\phi^2$$
$$- \left(1+\frac{\chi}{r}\right)^{\frac{\zeta_5\,\zeta_6}{\chi\,(\zeta_5+\zeta_6)}} a[t]^2\,p_4^2\,d\psi_4^2 - \left(1+\frac{\chi}{r}\right)^{-\frac{\zeta_5}{\chi}} a[t]^2\,p_5^2\,d\psi_5^2 - \left(1+\frac{\chi}{r}\right)^{-\frac{\zeta_6}{\chi}} a[t]^2\,p_6^2\,d\psi_6^2$$

Observation: Write the exponents as $e_\mu = -\frac{\zeta_\mu}{\chi}$, so that $\mathcal{D}_\mu(r) = (1+\frac{\chi}{r})^{e_\mu}$.

The two constraints $\chi = -(\zeta_4+\zeta_5+\zeta_6)$ and $\zeta_4\zeta_5 + \zeta_4\zeta_6 + \zeta_5\zeta_6 = 0$ then become $e_4 + e_5 + e_6 = 1$ and ${e_4}^2 + {e_5}^2 + {e_6}^2 = 1$.

These are exactly the Kasner conditions. They are the constraints on the exponents of the Kasner metric (1921),the classic vacuum solution of Einstein's equations in which three directions scale as different powers of time, $g_{\mu\mu}=(t^2)^{p_\mu}$. In 7D_1 the three directions are the curled circles, and $1+\frac{\chi}{r}$ plays the role that t^2 plays in Kasner's metric. Branch 1 is the Kasner triple with one exponent equal to 1 and the other two equal to 0, the distinguished circle being ψ_6. Branch 2 contains the rest of the Kasner circle, for example (2/3, 2/3, -1/3).

b: Solve $\mathcal{R}(t) = c^2$ to find a(t)

Solving $\frac{6\,\dot{a}[t]^2}{\Lambda\,a[t]^2} = c^2$ yields

$$a(t) = e^{\pm\frac{c\,t\,\sqrt{\Lambda}}{\sqrt{6}}}\,\alpha \quad \text{where } \alpha > 0 \text{ is a constant of integration and } \Lambda > 0$$

c: Kaluza-Klein (KK) bubble-of-nothing

Case 7D_1 has three curled circles, parameterized by ψ_4, ψ_5, and ψ_6, each with period 2π. $\mathcal{D}_4$, $\mathcal{D}_5$, and $\mathcal{D}_6$ modify p_4, p_5, and p_6, respectively, the radii of the three circles. They were not designed to be tied to r. However, through the GRW solution,

$$\mathcal{D}_\mu(r) = (1+\frac{\chi}{r})^{-\frac{\zeta_\mu}{\chi}},\ \mu=4,5,6,$$

all three circles developed an r-dependence. The proper radius of the ψ_μ circle is

$$R_\mu(t,r) = a(t)\,p_\mu\,\sqrt{\mathcal{D}_\mu[r]}\,.$$

The exponents Write $e_\mu = -\frac{\zeta_\mu}{\chi}$ for the three exponents. The two field-equation constraints from Section a become

$$\chi = -(\zeta_4 + \zeta_5 + \zeta_6) \quad \Leftrightarrow \quad e_4 + e_5 + e_6 = 1,$$

$$\zeta_4\,\zeta_5 + \zeta_4\,\zeta_6 + \zeta_5\,\zeta_6 = 0 \quad \Leftrightarrow \quad e_4^2 + e_5^2 + e_6^2 = 1,$$

These two conditions allow only two kinds of exponent triples:

- one exponent equals 1 and the other two equal 0; or
- exactly one exponent is negative and the other two lie strictly between 0 and 1 (example: $e = (2/3, 2/3, -1/3)$).

(Reason: if no exponent is negative, then $\Sigma e^2 \le (\Sigma e)^2 = 1$, with equality only when a single exponent is nonzero.)

$\chi > 0$ Then $1 + \frac{\chi}{r} > 0$ for every $r > 0$, so no $\mathbb{D}_\mu$ vanishes and no circle pinches off. There is no bubble and no KK constraint.

$\chi < 0$ Let $r_0 = -\chi > 0$. The solution is defined for $r > r_0$, and as $r \to r_0^+$ we have $1 + \frac{\chi}{r} \to 0$. So $B = \frac{1}{1 + \frac{\chi}{r}} \to \infty$, every circle with $e_\mu > 0$ shrinks to zero, a circle with $e_\mu = 0$ keeps its radius a p_μ, and a circle with $e_\mu < 0$ grows without bound.

Near r_0, the proper radial distance from the pinch is

$$\rho = \int a\,\sqrt{B}\,dr \approx 2a\,\sqrt{r_0\,(r - r_0)}, \quad \text{so} \quad r - r_0 \approx \frac{\rho^2}{4\,a^2\,r_0},$$

and the circle radii become

$$R_\mu \approx a\,p_\mu\,\left(\frac{r - r_0}{r_0}\right)^{\frac{e_\mu}{2}} = a\,p_\mu\,\left(\frac{\rho}{2\,a\,r_0}\right)^{e_\mu}.$$

Smoothness at the pinch (optional requirement) If we require the

geometry to close off smoothly at $r = r_0$, the way the origin of a flat plane does in polar coordinates, then each shrinking circle must satisfy $R_\mu \approx \rho$ as $\rho \to 0$:

- An exponent strictly between 0 and 1 is ruled out. The (ρ, ψ_μ) surface then has Gaussian curvature $-e_\mu \frac{e_\mu - 1}{\rho^2}$, which blows up at $\rho = 0$.
- A negative exponent is ruled out, because that circle blows up at the pinch.
- $e_\mu = 1$ works, provided $\frac{p_\mu}{2\, r_0} = 1$, i.e. $\chi = -\frac{p_\mu}{2}$. Otherwise the surface has a conical defect.

Combined with the exponent conditions above, smoothness therefore requires one exponent equal to 1 and the other two equal to 0. Taking ψ_6 as the pinching circle:

$$\zeta_4 = \zeta_5 = 0, \quad \zeta_6 = -\chi, \quad \chi = -\frac{p_6}{2},$$

$$\mathcal{D}_4 = \mathcal{D}_5 = 1, \quad \mathcal{D}_6(r) = 1 - \frac{p_6}{2\, r},$$

which is Branch 1 of Section a. (Choosing ψ_4 or ψ_5 as the pinching circle gives the same result with the indices permuted.) The factor $a(t)$ cancels from $\frac{R_\mu}{\rho}$, so the condition holds at every time.

p_4, p_5, and p_6 remain free constants: p_4 and p_5 because their circles never pinch, and p_6 as the one remaining modulus after smoothness fixes χ in terms of it. This is the same situation as 6D_1, where the ψ_5 circle pinched and smoothness gave $\chi = -\frac{p_5}{2}$.

Without the smoothness requirement The full two-parameter family from Section a stands as an exact solution of the Λ vacuum field equations. For $\chi < 0$ every member other than the special case above has a curvature singularity at $r = r_0$.

FG-GRW The FG-GRW solution is the GRW solution with $r\sqrt{C}$ in place of r throughout. In particular, $B = \frac{(2C + rC')^2}{4CQ}$ is exactly $\frac{\left[\left(r\sqrt{C}\right)'\right]^2}{Q}$.

So the pinch analysis is identical, with $r\sqrt{C}$ in place of r, and the KK-bubble constraint is the same, $\chi = -\frac{p_6}{2}$. Thus, one "KK-bubble Constraints" row serves both halves of the table, below, in Section d.

d: Free-gauge GRW

The table below shows the Branch 2 GRW solution, obtained in Section a, and the FG-GRW solution we are about to obtain. It also provides comparison to prior cases.

		5D	6D_1	6D_2	7D_1
GRW	Con-straints	None	$\Lambda>0,\ k=0$	$k=\frac{1}{p_4^2}>0$ $\chi=0$	$\Lambda>0,\ k=0$ $\chi=-(\zeta_4+\zeta_5+\zeta_6)\neq 0$ $\zeta_4=-\frac{\zeta_5\,\zeta_6}{\zeta_5+\zeta_6}$
	KK-bubble Con-straints (optional)	$\chi=r_0(-1+k r_0^2)$ $r_0=\frac{\mp 1+\sqrt{1+3\,k\,p_4^2}}{3\,k\,p_4}$ $k\geq -\frac{1}{3\,p_4^2}$	$\chi=-\frac{p_5}{2}$	None	$\zeta_4=\zeta_5=0$ $\chi=-\frac{p_6}{2}$
	$\mathcal{A}$	$\mathcal{A}(t)=c^2$	$\mathcal{A}(t)=c^2$	$\mathcal{A}(t)=c^2$	$\mathcal{A}(t)=c^2$
	B	$\frac{1}{\mathcal{D}_4}$	$\frac{1}{\mathcal{D}_5}$	$\frac{1}{\mathcal{D}_4}$	$\frac{1}{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6}$
	C	1	1	1	1
	$\mathcal{D}_4$	$1-k r^2+\frac{\chi}{r}$	1	$1-k r^2+\frac{\chi}{r^{2/3}}$	$\left(1+\frac{\chi}{r}\right)^{-\frac{\zeta_4}{\chi}}$
	$\mathcal{D}_5$		$1-kr^2+\frac{\chi}{r}$		$\left(1+\frac{\chi}{r}\right)^{-\frac{\zeta_5}{\chi}}$
	$\mathcal{D}_6$				$\left(1+\frac{\chi}{r}\right)^{-\frac{\zeta_6}{\chi}}$
FG GRW	$\mathcal{A}$	$\frac{4\,\dot{a}^2}{-4\,k+a^2\,\Lambda}$	$\frac{5\,\dot{a}^2}{-5\,k+a^2\,\Lambda}$	$\frac{5\,\dot{a}^2}{-5\,k+a^2\,\Lambda}$	$\frac{6\,\dot{a}^2}{-6\,k+a^2\,\Lambda}$
	B	$\frac{(2\,C+r\,C')^2}{4\,C\mathcal{D}_4}$	$\frac{(2\,C+r\,C')^2}{4\,C\mathcal{D}_5}$	$\frac{(2\,C+r\,C')^2}{4\,C\mathcal{D}_4}$	$\frac{(2\,C+r\,C')^2}{4\,C\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6}$
	C	C (free)	C (free)	C (free)	C (free)
	$\mathcal{D}_4$	$1-k r^2\,C+\frac{\chi}{r\sqrt{C}}$	1	$1-k r^2\,C+\frac{\chi}{r^{2/3}\sqrt[3]{C}}$	$\left(1+\frac{\chi}{r\sqrt{C}}\right)^{-\frac{\zeta_4}{\chi}}$
	$\mathcal{D}_5$		$1-k r^2\,C+\frac{\chi}{r\sqrt{C}}$		$\left(1+\frac{\chi}{r\sqrt{C}}\right)^{-\frac{\zeta_5}{\chi}}$
	$\mathcal{D}_6$				$\left(1+\frac{\chi}{r\sqrt{C}}\right)^{-\frac{\zeta_6}{\chi}}$

Notes:

- The 7D_1 column lists the Branch 2 solution of Section a explicitly.
- Branch 1 is the special case of $\zeta_5=0$, which gives $\zeta_4=0$ and $\chi=-\zeta_6$,

so that $\mathcal{D}_4=\mathcal{D}_5=1$ and $\mathcal{D}_6=1+\frac{\chi}{r}$.

✓ Though it is a special case, Branch 1 is the specific solution that results from accepting the KK bubble smoothness constraints.

- $\mathcal{A}(t) = c^2$ is the Friedmann Continuity Equation.
  ✓ The solution to the equation is the a(t) formula from Section b
- The two constraint rows display restrictions on parameter constants
  ✓ Due to the field equations
  ✓ Due to KK-bubble smoothness
- By contrast, domain restrictions on r (RCPC and its analogs) are not displayed as constraints
- FG-GRW has the same constraints (both rows) as GRW

The 7D_1 Free-gauge GRW results in the table were obtained from GRW by using the 6D_1 pattern:

$\mathcal{A}$: Replace c^2 by $\mathcal{A}$ from the $\mathcal{A}[t]$ formula

B: Use the formula in the table, where $\mathcal{D}_\mu$ = (FG-GRW $\mathcal{D}_\mu$)

C: Replace 1 by free parameter C

$\mathcal{D}_4$, $\mathcal{D}_5$, and $\mathcal{D}_6$: Replace r by $r\sqrt{C}$

The line element is

$$ds^2 = \frac{6\,\dot{a}[t]^2}{\Lambda\,a[t]^2}\,dt^2 - \frac{a[t]^2\left(1+\frac{\chi}{r\sqrt{C[r]}}\right)^{\frac{\zeta_5}{\chi}+\frac{\zeta_6}{\chi}-\frac{\zeta_5\zeta_6}{\chi(\zeta_5+\zeta_6)}}(2\,C[r]+r\,C'[r])^2}{4\,C[r]}\,dr^2 - r^2\,a[t]^2\,C[r]\,d\theta^2$$
$$- r^2\,a[t]^2\,\mathrm{Sin}[\theta]^2\,C[r]\,d\phi^2$$
$$- a[t]^2\,p_4^2\left(1+\frac{\chi}{r\sqrt{C[r]}}\right)^{\frac{\zeta_5\zeta_6}{\chi(\zeta_5+\zeta_6)}}d\psi_4^2 - a[t]^2\,p_5^2\left(1+\frac{\chi}{r\sqrt{C[r]}}\right)^{-\frac{\zeta_5}{\chi}}d\psi_5^2 - a[t]^2\,p_6^2\left(1+\frac{\chi}{r\sqrt{C[r]}}\right)^{-\frac{\zeta_6}{\chi}}d\psi_6^2$$

Confirming that FG-GRW satisfies all the field equations:

R_{00}: $\Lambda = \Lambda$

R_{01}: $0 = 0$

R_{02}: $0 = 0$

R_{03}: $0 = 0$

R_{04}: $0 = 0$

R_{05}: $0 = 0$

R_{06}: $0 = 0$

R_{11}: $\Lambda = \Lambda$

R_{12}: $0 = 0$

R_{13}: $0 = 0$

R_{14}: $0 = 0$

R_{15}: $0 = 0$

R_{16}: $0 = 0$

R_{22}: $\Lambda = \Lambda$

R_{23}: $0 = 0$

R_{24}: $0 = 0$

R_{25}: $0 = 0$

R_{26}: $0 = 0$

R_{33}: $\Lambda = \Lambda$

R_{34}: $0 = 0$

R_{35}: $0 = 0$

R_{36}: $0 = 0$

R_{44}: $\Lambda = \Lambda$

R_{45}: $0 = 0$

R_{46}: $0 = 0$

R_{55}: $\Lambda = \Lambda$

R_{56}: $0 = 0$

R_{66}: $\Lambda = \Lambda$

e: The Vacuum Friedmann Equations

Plugging $\mathcal{A}$, B, C, and D_4-D_6 (but not a) from the Branch 2 GRW solution into the field equations yields:

R_{00}: $\frac{6\,\ddot{a}}{a\,c^2} = \Lambda$

R_{11}: $\frac{5\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{22}: $\frac{5\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{33}: $\frac{5\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{44}: $\frac{5\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{55}: $\frac{5\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{66}: $\frac{5\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{22}-R_{66} is redundant, the same as R_{11}.

The Friedmann continuity and acceleration equations, respectively, are the following pair of equations generated by algebraic manipulation of R_{00} and R_{11}:

$$-\frac{1}{30}R_{00}+\frac{1}{5}R_{11}: \quad \frac{\dot{a}^2}{a^2\,c^2} = \frac{\Lambda}{6}$$

$$\frac{1}{6}R_{00}: \quad \frac{\ddot{a}}{a\,c^2} = \frac{\Lambda}{6}$$

It is easy to see that $\mathcal{R}(t) = \frac{6\,\dot{a}^2}{a^2\,\Lambda} = c^2$ is simply a rearrangement of the continuity equation.